MILITARY SPECIFICATION

HOOK, THRUST, JATO - 1000 POUNDS NOMINAL THRUST

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 This specification establishes the manufacture, inspection, and testing requirements for hook, thrust, JATO - 1000 pounds nominal thrust.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on the date of invitation for bids, forms a part of this specification.

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
PPP-C-843	Cushioning Material, Cellulose
PPP-B-601	Boxes, Wood, Cleated, Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-676	Boxes, Set-Up, Paperboard
UU-P-271	Paper, Wrapping, Waterproofed Kraft

Military

MIL-D-5028	Drawing and Data Lists; Preparation of (for Engines, Accessories, and other Auxiliary Equipment)
MIL-A-140	Adhesives, Water-Resistant, Waterproof Barrier Material
MIL-J-8667 (Aer)	Jet-Assisted Take-Off Units - Installation of in Naval Aircraft
MIL-V-7173	Vibration Tests of Fuel, Oil, and Miscellaneous Tanks; Process for
MIL-F-7179	Finishes and Coatings; General Specifications for Protection of Aircraft and Aircraft Parts
JAN-P-100	Packaging and Packing for Overseas Shipment; General Specifications
JAN-P-108	Packaging and Packing for Overseas Shipment; Boxes, Fibreboard (V-Board and W-Board) Exterior and Interior

STANDARDS

Federal Test Method - Std. 151	Metals; Test Methods
MIL-STD-129	Marking of Shipment and Storage

DRAWINGS

(Sheets 1 and 2) NAF400095	Hook, Thrust - JATO; 1000 pounds Nominal Thrust
NAF49489	Jig

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PUBLICATIONS

Air Force - Navy Aeronautical Bulletin No. 143 Specifications and Standards; Use of Navy Department - General Specification for the Inspection of Materials

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code AD.)

3. REQUIREMENTS

3.1 Qualification.- The subject thrust hooks furnished under this specification shall be a product which has passed the preproduction and inspection tests specified herein.

3.2 Materials.- Material shall conform to the applicable specifications as specified herein and on the applicable drawings. Materials which are not covered by applicable specifications shall be of the best commercial quality, of the lightest practical weight, and entirely suitable for the purpose intended.

3.2.1 Selection of Materials.- Specifications and Standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.1.1 AN or MIL Standard Parts.- AN or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they possess suitable properties and are replaceable by the AN or MIL Standard Parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or MIL Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.4 Performance.- Hooks shall function properly under all tests specified under Section 4.

3.5 Workmanship.- The workmanship shall be that required by the best commercial practice, covering this type equipment, for the quantity production of analogous parts. All operating parts of the hooks shall be free from imperfections which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of hooks shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the item after the award of contract, to determine that the production meets the requirements of this specification.

(b) Inspection tests: Inspection tests are those tests accomplished on hooks, manufactured and submitted for acceptance under contracts.

4.2 Inspection of Materials:

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4.2.1 Commercial Materials.- When materials are procured under commercially recognized non-governmental specifications, such as SAE numbers, a certificate showing compliance with such specifications shall be furnished by the supplier of the material. Such a certificate shall be in a form acceptable to the Government Inspector.

4.2.2 Approval of Materials in Process.- Inspection and approval of materials in process during the course of manufacture shall not be construed as insuring the acceptance or approval of the finished product.

4.3 Preproduction Tests:

4.3.1 Sampling Instructions.- The first three hooks produced under each contract, and fabricated from production tools, shall be preproduction test samples and shall be delivered for inspection and testing as directed by the Procuring Service, together with a certification by the cognizant inspector, that the materials used in their manufacture are in accordance with pertinent specifications.

4.3.2 Tests.- The Preproduction Test samples shall be subject to inspection for conformance with pertinent drawings and specifications indicated in Section 2, and the following tests, as described under "Test Methods" which will be conducted by a Testing Activity under the direction of the Procuring Service.

TITLE	APPLICABLE PARAGRAPH	SAMPLE #1	SAMPLE #2	SAMPLE #3
Physical Characteristics	4.5.1		All	
Salt Spray Test	4.5.2			X
Vibration Tests	4.5.3	X		
Release Function Test	4.5.4		All	
Life Test	4.5.5		X	X
Static, Repeat Load and Destruction Test	4.5.6, 4.5.7	X		X

4.3.3 Correction of Defects.- Any defects discovered during the inspection and testing of the Preproduction Test samples shall be corrected by the contractor. Such measures as may be found necessary to correct these defects shall likewise be incorporated in all subsequent hooks offered for acceptance under contract.

4.4 Inspection Tests:

4.4.1 General.- The complete hooks, as described and defined herein, together with all materials and processes used in the manufacture thereof, shall be subject to inspection at the contractor's plant, or elsewhere, by authorized Government Inspectors, in general accordance with the provisions of the Navy Department Specification for the Inspection of Materials. Inspection tests shall be conducted by a testing activity to be designated by the Procuring Service in accordance with the sampling technique and methods of testing specified herein.

4.4.2 Sampling Instructions.- The following number of production units shall be selected at random from each production lot:

No. of Hooks per Lot	No. of Samples Selected
99 or less	2
100 to 299	3
300 to 500	4

NOTE: When the lot size exceeds 500 hooks, the lot shall be subdivided into lots of 500 units and the selection for each subdivision shall be in accordance with the above table.

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4.4.3 Tests.- Samples for the Inspection Tests, selected in accordance with the preceding paragraphs, shall in addition to Examination of Product and Magnaflux inspection, be subjected to the following tests, which will be conducted by a Test Activity:

Physical Characteristics	All of the samples shall be subjected to this test.
Release Function Test	All of the hooks shall be subjected to this test.
Life Test	All of the samples shall be subjected to this test.
Static, Repeat Load, and Destruction Test	All of the samples shall be subjected to this test.

4.4.4 The following additional tests shall be conducted at the contractor's plant:

4.4.4.1 Examination of Product.- Each hook submitted for acceptance under contract shall be inspected visually to insure that it has been properly assembled and that the workmanship and the finishes are in accordance with the provisions of this specification.

4.4.4.2 Magnaflux Inspection.- After machining and prior to plating, each hook shall be subjected to a magnaflux inspection. The magnaflux inspection shall be repeated after plating. Inspection certificates shall be made available to the Government Inspectors.

4.4.5 Rejection and Retest.-

4.4.5.1 Inspection Test Samples.- The failure of any Inspection Test sample to pass the inspections and tests outlined in paragraph 4.4.3 shall cause tentative rejection of all the hooks represented by that sample, subject to retest as hereinafter provided.

4.4.5.1.1 First Retest.- In the event any sample representing a production lot fails to pass the inspection and tests outlined in paragraph 4.4.3, the contractor shall have the opportunity of reworking the entire affected lot and resubmitting to test, twice as many samples, selected at random by the Government Inspector, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.4.5.1.2 Second Retest.- If, in the contractor's opinion, the failure of the retest samples is due to a remediable defect, the contractor may, at the discretion of the inspector, be offered the opportunity of reworking the entire affected lot and resubmitting to test three times as many samples, to be selected at random by the Government Inspector, as originally submitted for the entire lot. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.4.5.2 Individual Inspection Samples.- Any hook or component part of a hook which fails to pass the individual inspection and tests as outlined in paragraph 4.4.4 may, upon the request of the contractor, be resubmitted for individual inspection and test, providing a statement showing corrective measures taken, is furnished to the Inspector at the time of such resubmission. Upon reinspection and retest, the failure to meet any of the requirements outlined in paragraph 4.4.4 shall be cause for rejection.

4.4.5.3 Salvaged Pieces.- Any pieces from a rejected hook may be salvaged for use providing each piece is inspected for, and is in complete compliance with, the pertinent drawings and specifications.

4.4.5.4 Inspection Subsequent to Delivery.- Any or all hooks subsequent to the first three special test samples, may be subject to inspection and testing for conformance with any or all the requirements of this specification, after delivery to the Government.

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4.5 Test Methods:

4.5.1 Physical Characteristics.- The units submitted for this test shall be photographed if necessary to illustrate clearly appearance and general arrangement. The units shall also be weighed and subjected to a dimensional, material, and a process check.

4.5.2 Salt Spray Test.- Salt Spray Test shall be made in accordance with Federal Specification QQ-M-151.

4.5.3 Vibration Test.- Test hooks shall be mounted in a slosh vibration machine so that the longitudinal axis of mockup Jato bottle is at right angle to the axis of rotation of slosh machine.

Jato hooks shall be vibrated twenty-five (25) hours as follows:

1. Vibration frequency - 1700 - 2400 CPM
2. Vibration amplitude - .032"
3. Slosh frequency - 19 CPM, through a total angle of 30°, approximately 15° on each side of the level position.

Any undue wear of hook components or release of bottle during test period due to vibration, mechanical failure, or loosening of parts shall be considered cause for rejection of test hook.

4.5.4 Release Function Test.- Manufacturer shall prepare a test apparatus suitable for simulating the jettisoning of a 141# mock jato bottle. The hooks shall be so mounted on the apparatus that the force required to jettison the unit shall not exceed those forces outlined in paragraph 3.3.5.4.1 of Specifications MIL-J-8667 (AER) for Type I Jato Hooks.

Ten consecutive operations resulting in complete jettisoning of the bottle shall comprise a successful test. Each unit manufactured shall be so tested.

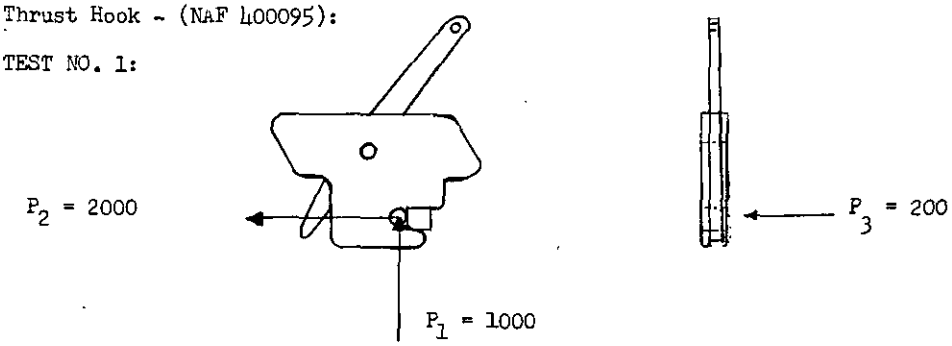
4.5.5 Life Test.- The life test fixture shall be constructed such that the test hooks are held in a horizontal position. The test hooks shall then be subjected to 500 releases. The test shall be conducted at room temperature and a mock jato bottle of 141# used throughout the test. Record the efficiency and percentage of total retraction between the 1st and 10th release, inclusive, and the 491st to 500th, inclusive.

4.5.6 Static and Repeat Load Tests:

4.5.6.1 Static tests shall be conducted on the sample hooks with loads shown applied simultaneously as follows:

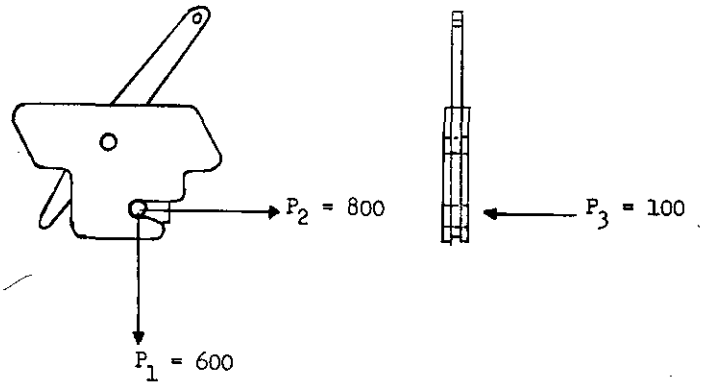
4.5.6.1.1 Thrust Hook - (NAF 400095):

a. TEST NO. 1:



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b. TEST NO. 2:

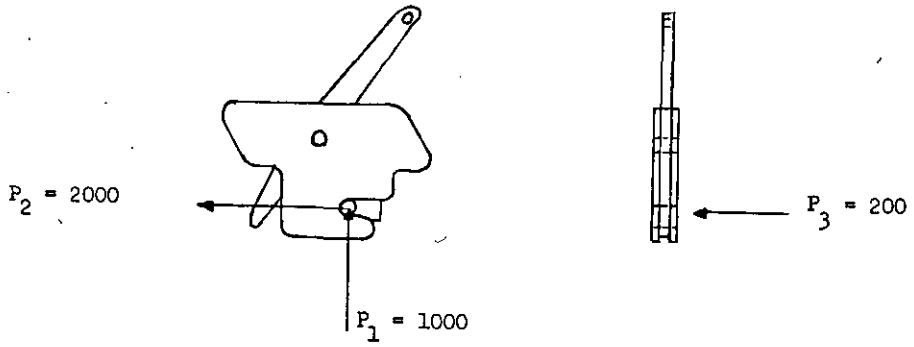


The above loads are limit loads. Tests shall be run to proof loads (1.10 x limit load) and there shall be no permanent deformation or malfunctioning of any part of the hook below the proof load.

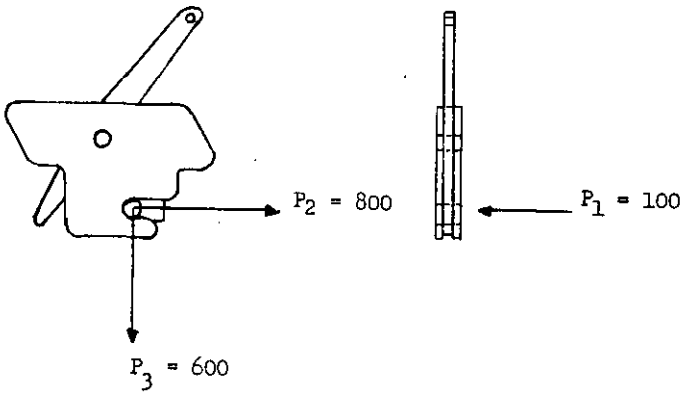
4.5.6.2 Repeat load tests shall be conducted on the sample hook as follows:

4.5.6.2.1 Thrust Hook - (NAF 400095):

a. TEST NO. 3:



b. TEST NO. 4:



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The repeat load tests are cycled from zero to the specified load and back to zero. There shall be no permanent deformation or malfunctioning of parts below 2000 cycles.

4.5.7 Destruction Tests:

4.5.7.1 Static Destruction Tests:

4.5.7.1.1 Thrust Hook - (NAF 400095):

a. Loads shall be applied in the same manner as indicated in 4.5.6.1.1 a and 4.5.6.1.1 b. Failure shall not occur below values of:

P_2 = 3000 lbs. P_1 = 1500 P_3 = 300

4.5.7.2 Repeat Load Destruction Tests:

4.5.7.2.1 Thrust Hook - (NAF 400095):

a. Magnitude and direction of load to be as shown in 4.5.6.2.1 a and 4.5.6.2.1 b. Failure shall not occur below 30,000 cycles.

5. PREPARATION FOR DELIVERY

5.1 Application.- The preservation, packaging, packing, and marking requirements specified herein apply only to direct purchases or by direct shipment to the Government.

5.2 Preservation.- Each hook shall be preserved in accordance with the requirements of method 1A, of Specification AN-P-13, without the use of preservative compound.

5.3 Packaging.- Unless otherwise specified, cushioning shall be provided to prevent rupture of the method 1A barrier, using materials set forth in Specifications PPP-C-843 and JAN-P-100. Unit containers shall be folding cartons, set-up boxes, or fibreboard boxes, conforming to Specifications PPP-B-566, PPP-B-676, or JAN-P-108, and each container shall contain one complete hook (preserved) which shall be snugly packaged.

5.4 Packing.- Unless otherwise specified, all hooks shall be afforded domestic packing. Shipping containers shall contain the same number of unit packages, shall be uniform in size, and snugly packed. The gross weight, when packed for shipment, shall not exceed approximately 200 pounds.

5.4.1 Domestic Packing.- Unless otherwise specified, the unit packages shall be placed in substantial shipping containers so constructed as to insure acceptance by common or other carrier for safe transportation at lowest rates to the point of delivery specified in the contract or order. Containers shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment, except that fibreboard, when used, shall have a minimum Mullen test of 275 pounds.

5.4.2 Overseas Packing.- Unless otherwise specified, the unit packages shall be in cleated plywood, or nailed wood, boxes conforming to Specifications PPP-B-601 or PPP-B-621. Plywood, when used, shall be type A or B, condition 1, Specification MN-P-515. Boxes shall be furnished with a sealed case liner fabricated from case liner material, Specification MIL-P-140.

5.5 Marking and Labeling.-

5.5.1 Packages.- Each unit package shall be durably and legibly marked or labeled with the following information:

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Jato Hook - Thrust
1000# Nominal Thrust
Specification MIL-
Preserved (date):
Domestic or Overseas Packed (as applicable)
Contract or Order No.
Name of Manufacturer
Name of Contractor (if different from manufacturer)

5.5.1.1 In addition, each container shall be marked in accordance with the requirements applicable to the individual Services, as specified in MIL. Standard 129.

6. NOTES:

6.1 Adherence to Drawings and Specifications: Conflict Elimination - Insofar as the contractor is required by the provisions of this specification to adhere rigidly to the drawings listed in paragraph 2, which may make it impossible in some instances to also comply with the performance and test requirements specified herein, then the contractor should promptly notify the Procuring Service of such facts and appropriate changes will be made in the applicable drawings, parts or performance requirements by the Procuring Service. An equitable adjustment in accordance with applicable contract provisions will be made in the price of the articles to be delivered to reflect any change so authorized by the Bureau of Aeronautics.

6.2 Information Required with Bids.- Each bid submitted in connection with the procurement of the hooks shall be accompanied by a statement showing whether or not any exceptions are being taken to the requirements shown in the drawings or in the specification. If exceptions are taken, they shall be specifically described, and the alternate thereto indicated in the statement.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in anyway be related thereto.

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